

Kernel Composition Summary

6 February 2026

Overview

- ▶ Summarizes kernel composition experiments.
- ▶ Compares composition runs against prior MMR/ABESS baselines.
- ▶ Highlights best-performing configuration and key visuals.

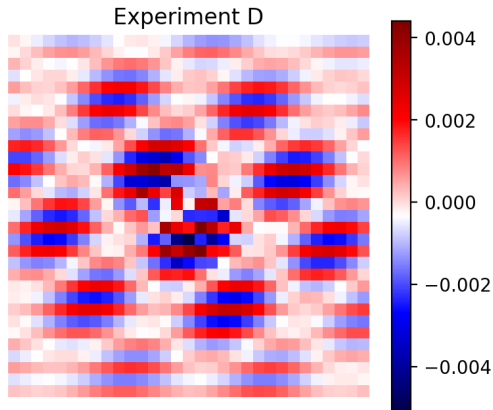
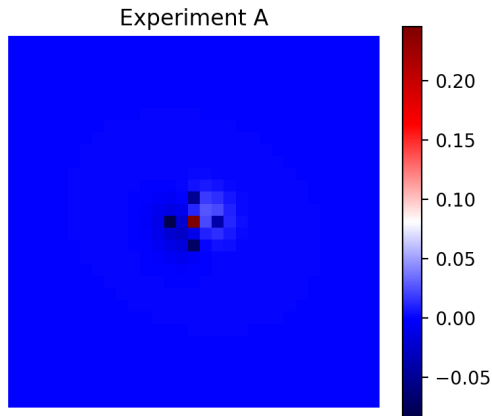
This deck focuses on how composition size and per-family bank size affect test AUC/ACC.

Composition Results

Run	Test AUC	Test ACC	Comp. pool
A	0.8662	0.7968	25
B	0.8666	0.7920	125
C	0.8869	0.8107	625
D	0.8933	0.8150	1250

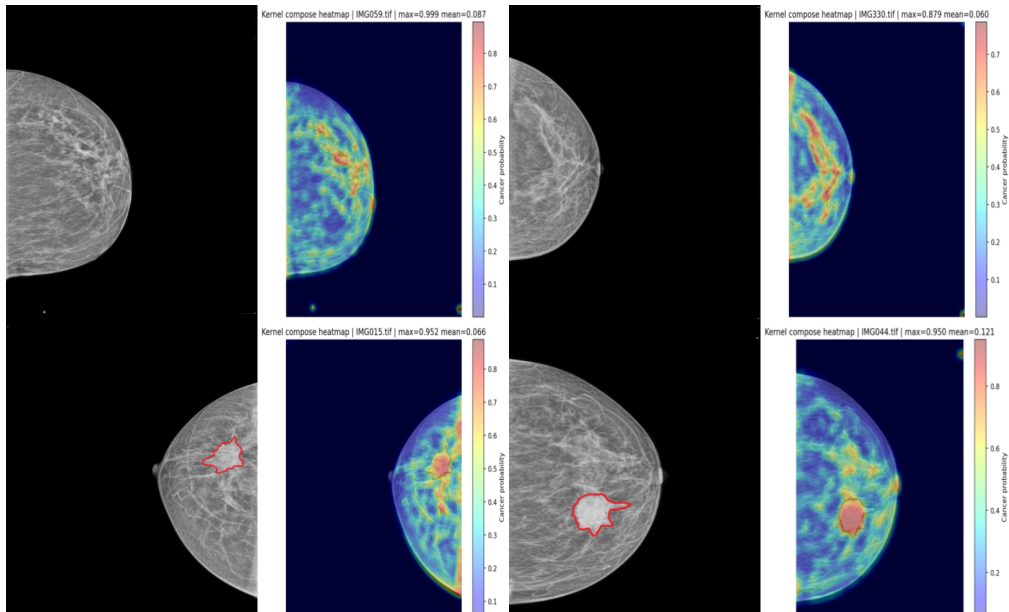
Larger composition pools improve composite test AUC/ACC; D is strongest.

Composite Kernel Visualization



Red/blue indicate positive/negative response regions; stronger contrast suggests a more selective filter.

Heatmap Example (Best Composition)



How parameter changes affect quality

- ▶ Increasing composition size ($25 \rightarrow 125$) improves both test AUC and test accuracy.
- ▶ Increasing per-family kernel count ($10 \rightarrow 50$) yields a stronger composite filter and better generalization.
- ▶ Smaller banks keep accuracy competitive but usually reduce AUC.
- ▶ Best quality comes from a larger kernel bank and a larger composition size.

Composition vs non-composition accuracy

Run	Test ACC (regular)	Test ACC (composite)
A	0.8487	0.7968
B	0.8503	0.7920
C	0.8492	0.8107
D	0.8733	0.8150

Regular models score higher, while composites provide a stable, interpretable baseline.

Conclusion

- ▶ Best overall: run D (500 per-family, 1250 composition).
- ▶ Highest composite test AUC (0.8933) and composite test ACC (0.8150).
- ▶ It also delivers the best regular-model test accuracy (0.8733).

Next Steps

- ▶ Run more experiments with different parameter settings to improve accuracy.
- ▶ Test on a new dataset to confirm generalization.
- ▶ Study a learned kernel-generating model to avoid random kernel sampling.

Recommendation

- ▶ Prefer a single composite kernel over a large uncombined kernel bank.
- ▶ It performs better in prior runs while being simpler and faster to deploy.
- ▶ Keep the large multi-kernel model only as a reference baseline.